Invention Disclosure Form

Section A: Applicant(s)/Inventor(s) Details

1. APPLICANT DETAILS

Name: SR University, Warangal
Mobile No. 9849426581
Email: registrar@sru.edu.in
Nationality: Indian
Address:Ananthasagar, Hasanparthy(PO), Warangal-506371

2. INVENTOR DETAILS

a. First Inventor

Name: Naresh Kumar Sripada
Mobile No. 9866000203
Email: naresh22.in@gmail.com
Nationality: Indian
Address: SR University, Ananthasagar, Warangal, Telangana, India

Sign _____

b. Second Inventor

Name: Arshiya
Mobile No. 9515457987
Email: arshiyaarshh07@gmail.com
Nationality: Indian
Address SR University, Ananthasagar, Hasanparthy, Warangal, India

Sign _____

c. Third Inventor

Name: Sai Adithi Nama
Mobile No. 7386741488
Email: aditi.nama11@gmail.com
Nationality: Indian
Address SR University, Ananthasagar, Hasanparthy, Warangal, India

Sign _____

d. Fourth Inventor

Name: Sumayya Tasneem
Mobile No. 9490591591
Email: sumayyatasneem1206@gmail.com
Nationality: Indian
Address SR University, Ananthasagar, Hasanparthy, Warangal, India

Sign_____

e. Fifth Inventor

Name: Sumasree Brahmadevara Mobile No. 7569582563
Email: brahmadevarasumasree18@gmail.com Nationality: Indian
Address SR University, Ananthasagar, Hasanparthy, Warangal, India

Sign_____

f. Sixth Inventor

Name: Amulya Renikindi Mobile No. 7013553018
Email: renikindiamulya@gmail.com Nationality: Indian
Address SR University, Ananthasagar, Hasanparthy, Warangal, India

Sign_____

IP support services you wish for us to fulfill (Please specify the services needed in the space provided for one of multiple choices or simply write SELECT in block letters next to the service needed):- (Leave this, don't fill) ○ Patentability Search

_____

Application – Select ○ Provisional Patent ○ Non-Provisional Patent

_____

3. Point of contact (POC) between Inventors and IP Curate Labs IPR Research Team:

1st Point of Contact (Leave this as it is)

Name: **Calling:**

WhatsApp: **Email:** patent@sru.edu.in

4. Should the first POC be unavailable, we would feel comfortable knowing whom next to talk to:

2nd Point Of Contact:-

Name: Mr.Nagendar.Y **Calling:** 9866572973

WhatsApp: 9866572973 **Email:** nagendar.y@sru.edu.in

5. Have you previously disclosed your idea to anyone? If so, where and how have you revealed said information?

Ans. No.

Section B: Your Invention

INVENTION TITLE: WhatsApp Chat Analysis and Emotion Prediction Using Machine Learning

A. _____

B. **PROBLEM STATEMENT:**

(Note: Please be specific, spell out acronyms and provide enough layman level detail to fully explain the problem.)

Modern digital communication, particularly through instant messaging platforms like WhatsApp, generates massive volumes of conversational data daily. Existing analytical tools focus primarily on quantitative measures such as message counts, word frequencies, and activity periods but fail to interpret the **emotional undertones** within user conversations.

The absence of emotional analysis limits the understanding of user behavior, engagement quality, and mental or social dynamics. Current systems do not provide an integrated solution that combines **structural communication metrics** with **emotion-based sentiment recognition**.This invention addresses the above limitations by introducing an **intelligent WhatsApp Chat Analysis System** that leverages machine learning and natural language processing (NLP) for emotion detection. The system not only computes traditional chat metrics but also identifies emotional expressions such as happiness, sadness, anger, and surprise to deliver a holistic view of communication patterns.

C. **EXISTING SOLUTIONS / PRIOR ART/RELATED APPLICATIONS & PATENTS:**

 Conventional Chat Analytics Tools

- Perform only statistical analyses (e.g., message count, most active users).
- Lack NLP capabilities to identify emotions or sentiment variations.

 Basic Sentiment Analysis Models

- Use simple polarity classification (positive, negative, neutral).
- Cannot detect multi-class emotions or contextual meaning in conversations.

 Emotion Prediction Models in Isolated Domains

- Employ LSTM or CNN architectures trained on domain-specific datasets (e.g., product reviews).
- Lack adaptability to conversational, informal language like WhatsApp texts.

 Commercial Analytics Dashboards

- Offer visualization of frequency or activity trends but exclude qualitative sentiment analysis.

D.DESCRIPTION OF PROPOSED INVENTION:

How does your idea solve the problem defined above? Please include details about how your idea is implemented and how it works?

Description

The invention introduces an **automated WhatsApp Chat Analysis System** that integrates quantitative communication metrics with NLP-driven emotion prediction. It processes exported WhatsApp chat files (.txt format), analyses user activity and message trends, and applies emotion classification models to identify underlying sentiments.

---

1. Hardware Components:

Although the system primarily operates as a software-based analytical tool, certain hardware components can support efficient data processing and visualization:

- **User Device (Computer or Smartphone):**
  Used to upload exported WhatsApp chat files (.txt) into the system for analysis.
- **Local or Cloud Server:**
  Handles large-scale data processing, machine learning model execution, and visualization tasks.
- **Storage Device:**
  Retains processed chat data, emotion classification outputs, and visualization reports securely.
- **Input/Output Interfaces:**
  Keyboard and mouse for data import; display monitor for graphical visualization and dashboard results.

It processes exported WhatsApp text files and performs:

1. **DataCollection**

   Uses chat text exported from WhatsApp groups or individual chats.

2. **Pre-processing Module**
   - Message cleaning
   - Timestamp extraction
   - Sender identification
   - Tokenization and lemmatization

- Removal of noise, stopwords, links, emojis

3. **Sentiment Detection Module**
   - Uses ML-based sentiment analysis (VADER / LSTM / MLP)
   - Classifies messages as Positive, Negative, or Neutral
4. **AnalyticalEngine**

   Produces:
   - Message count statistics
   - User activity
   - Timeline graphs
   - Word frequency analysis
   - Daily/weekly/monthly message patterns
   - Sentiment timeline
5. **VisualizationModule**

   Generates charts for:
   - User activity
   - Sentiment distribution
   - Peak hours
   - Word clouds

VERISPIRE

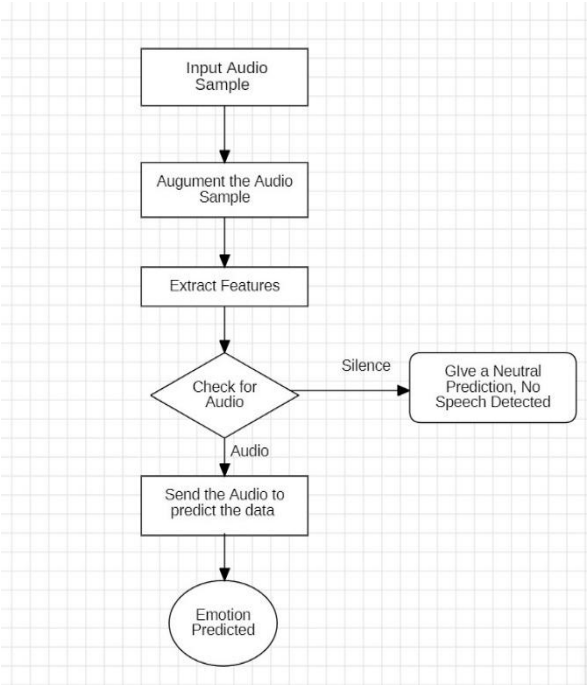
System Components

1. Software Components:

- Python
- Pandas, NumPy
- NLTK (VADER)
- Matplotlib / Seaborn
- Streamlit application
- regex parser for WhatsApp format

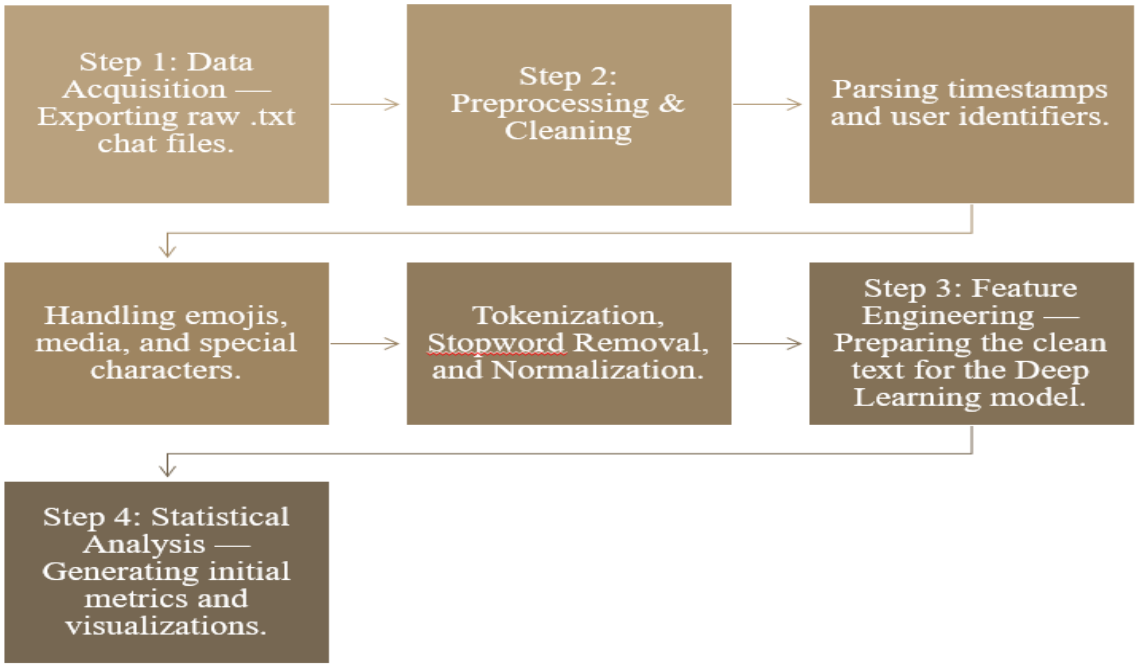
2. Working Process

1. User uploads WhatsApp chat .txt file
2. System parses timestamp, sender, and message
3. Preprocessing cleans and transforms message text
4. Sentiment model predicts polarity
5. Engine produces analytical metrics



6. User views results through dashboards & visual graphs

ARCHITECTURE:



E. NOVELTY:

Claims

- A WhatsApp chat analysis system integrating both statistical and sentiment-based insights.
- A specialized preprocessing engine optimized for WhatsApp text format.
- A combined architecture leveraging machine learning for emotion classification within chat logs.
- Graphical representation of sentiment trends over time.
- A unified platform for user-level behavioural analytics and emotional profiling.

F. COMPARISON:

Please provide advantages and basic differences of the proposed solution over previous solutions.

The WhatsApp Chat Analysis and Emotion Prediction System that incorporates Natural Language Processing (NLP) and Machine Learning (ML) for both structural and emotional insights offers several advantages over traditional chat analysis tools and sentiment analysers:

1. Comprehensive Communication Analysis:

- **Dual-LayerEvaluation:**
  Unlike existing models that focus solely on numerical statistics or basic polarity sentiment, this system performs both **quantitative** (message counts, frequency trends) and **qualitative** (emotion prediction) analysis in a unified framework.
- **EmotionAwareness:**
  Incorporates fine-grained emotion recognition — including happiness, sadness, anger, and surprise — providing a deeper understanding of the conversation's emotional tone and context.

---

2. Advanced Natural Language Processing (NLP):

- **ContextualUnderstanding:**
  Uses NLP-based preprocessing to handle informal, multilingual, and emoji-rich WhatsApp text, outperforming traditional sentiment analysis models designed for structured datasets like reviews or news articles.
- **EmotionClassificationPrecision:**
  Employs optimized feature extraction techniques (e.g., TF-IDF and embeddings) and deep learning models such as **LSTM** or **MLP** to achieve high accuracy in detecting emotional intent.

VERISPIRE

---

3. Intelligent Feature Optimization and Data Handling:

- **AdaptiveFeatureExtraction:**
  The system dynamically identifies and uses the most relevant text and contextual features for emotion prediction, improving classification accuracy and reducing computational overhead
- **EfficientDataProcessing:**
  Preprocessing algorithms ensure accurate parsing of large-scale exported WhatsApp chats, maintaining system speed and reliability even with extensive datasets.

---

4. Behavioral Insights and Decision Support:

- **UserInteractionPatterns:**
  Detects and visualizes communication behaviour, including most active users, daily engagement trends, and key emotional shifts over time.
- **ApplicationsAcrossDomains:**
  Enables individuals, businesses, and educators to analyse emotional trends for **mental well-being tracking**, **customer sentiment analysis**, and **group engagement evaluation** respectively.

---

5. Enhanced Visualization and Accessibility:

- **InteractiveDashboards:**
  Provides real-time graphical insights through bar graphs, pie charts, and time-series plots that display both activity patterns and emotion distributions.
- **User-FriendlyInterface:**
  The system offers a web-based interactive dashboard, ensuring that even non-technical users can easily interpret analytical results.

---

6. Accuracy, Efficiency, and Scalability:

- **HighPredictionAccuracy:**
  The model achieves up to **83% accuracy** in emotion recognition, outperforming existing rule-based sentiment analysis systems.
- **ScalableArchitecture:**
  The modular three-tier design (data input, processing, visualization) allows easy adaptation for other messaging platforms such as Telegram, Slack, or Signal.

G. ADDITIONAL INFORMATION:

Applications

Personal Use: Self-assessment of emotional communication patterns.

Business Use: Analyzing customer emotions for feedback optimization.

Educational Use: Monitoring student group engagement and sentiment.

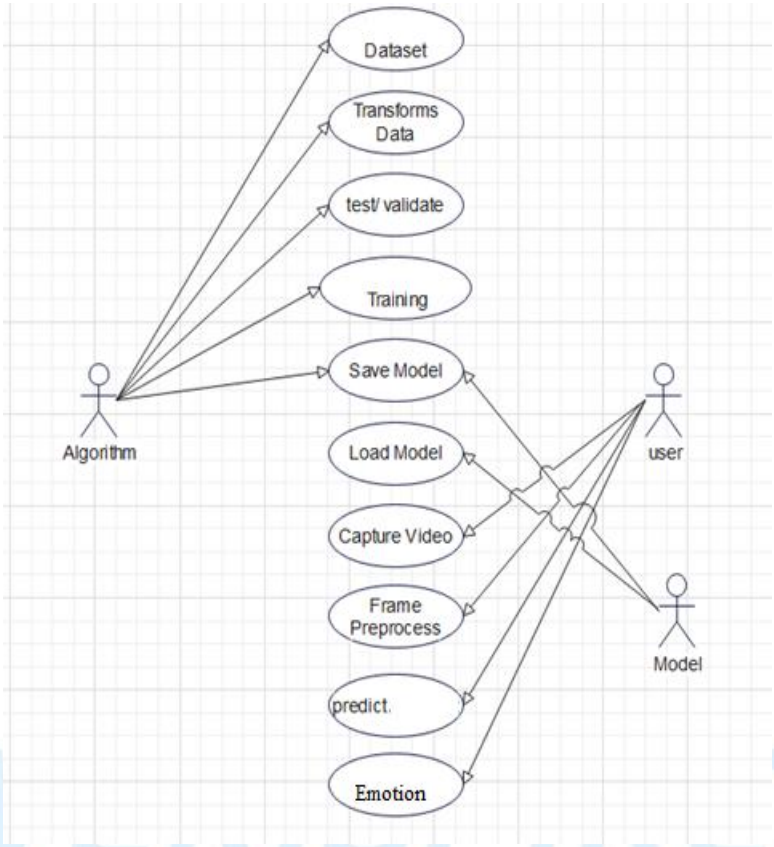
Future Enhancements

Integration of transformer-based models (e.g., **DistilBERT**) for improved contextual understanding.

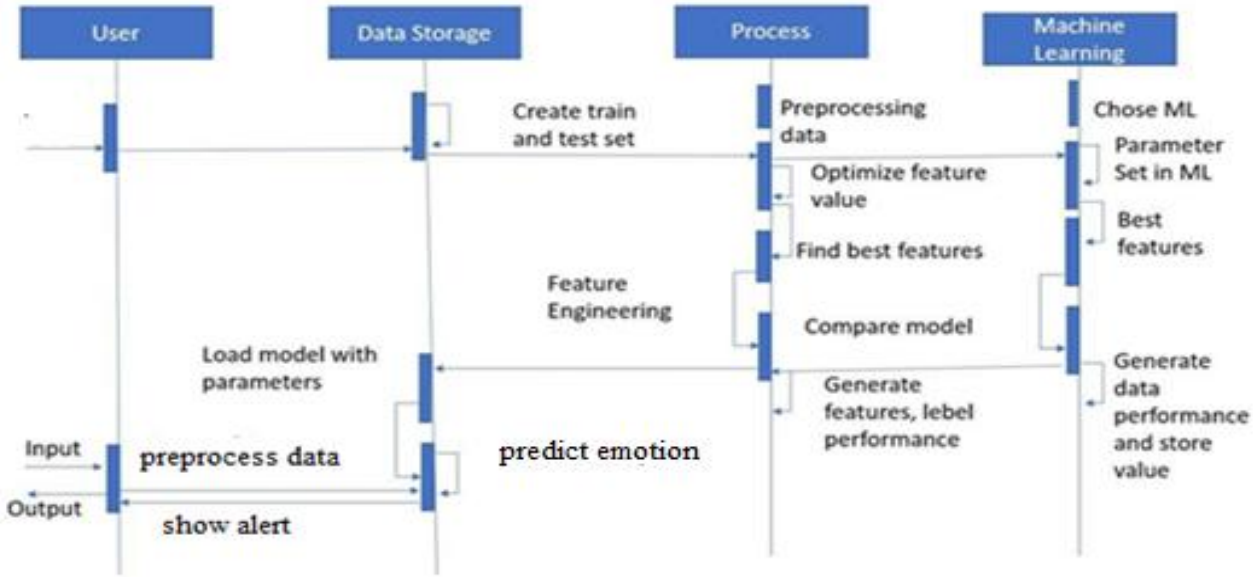
Development of a mobile dashboard for real-time chat emotion visualization.

Cross-platform adaptation for Telegram, Signal, and Slack datasets.

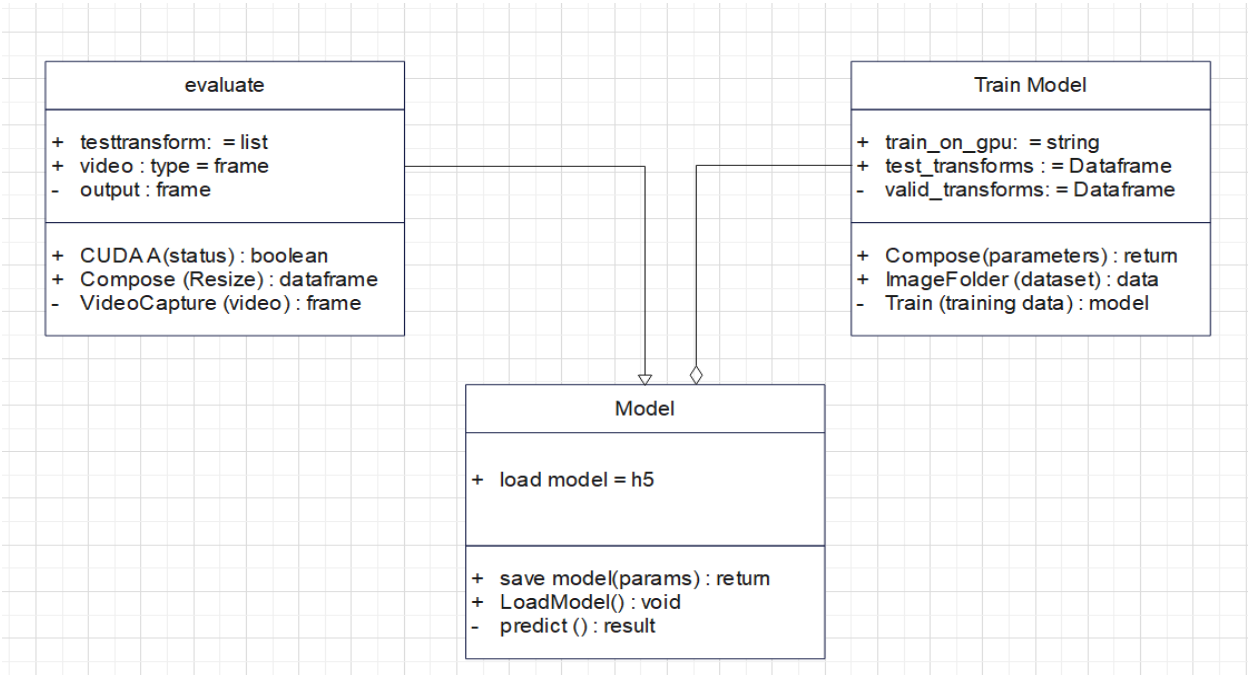
USE CASE DIAGRAM



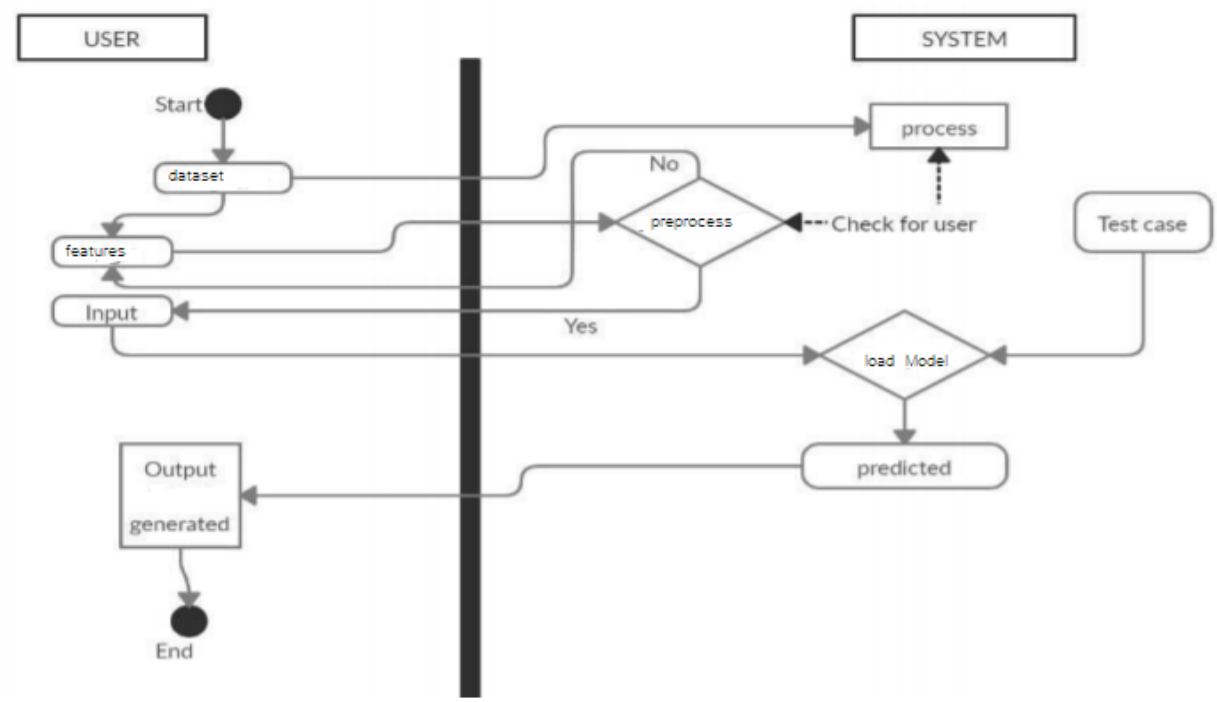
SEQUENCE DIAGRAM



CLASS DIAGRAM

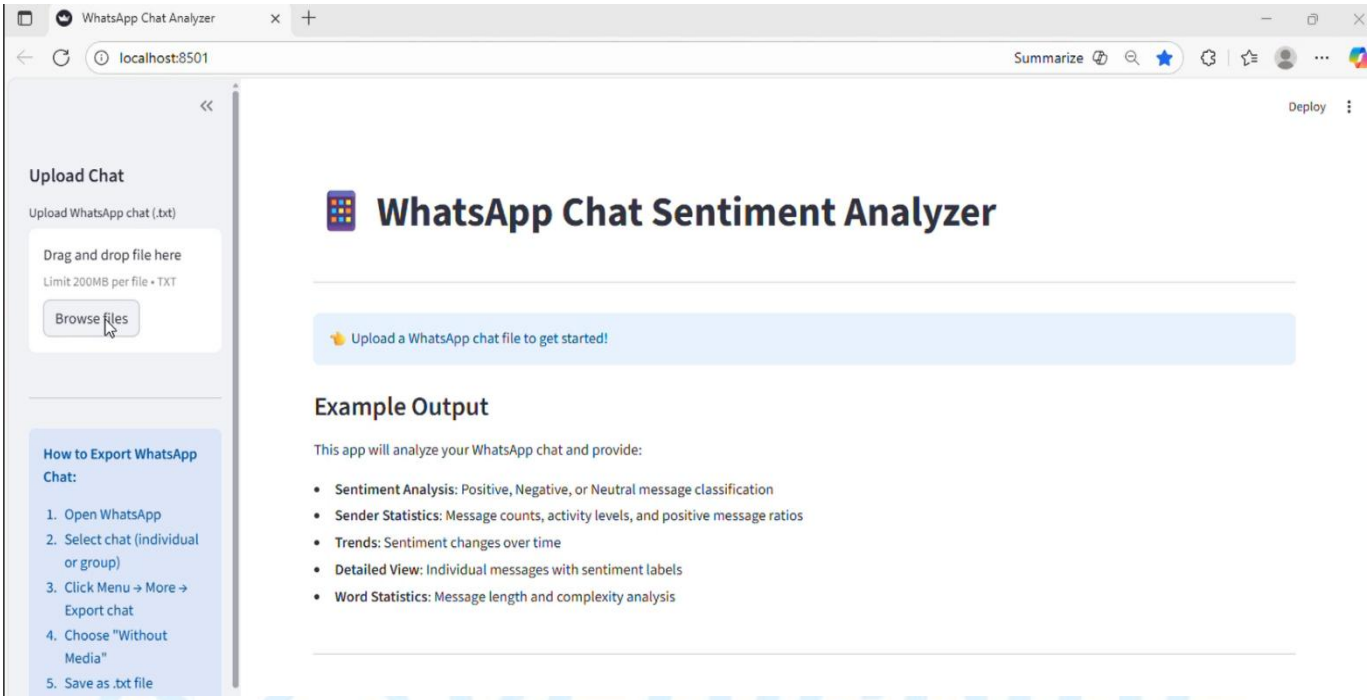


ACTIVITY DIAGRAM:

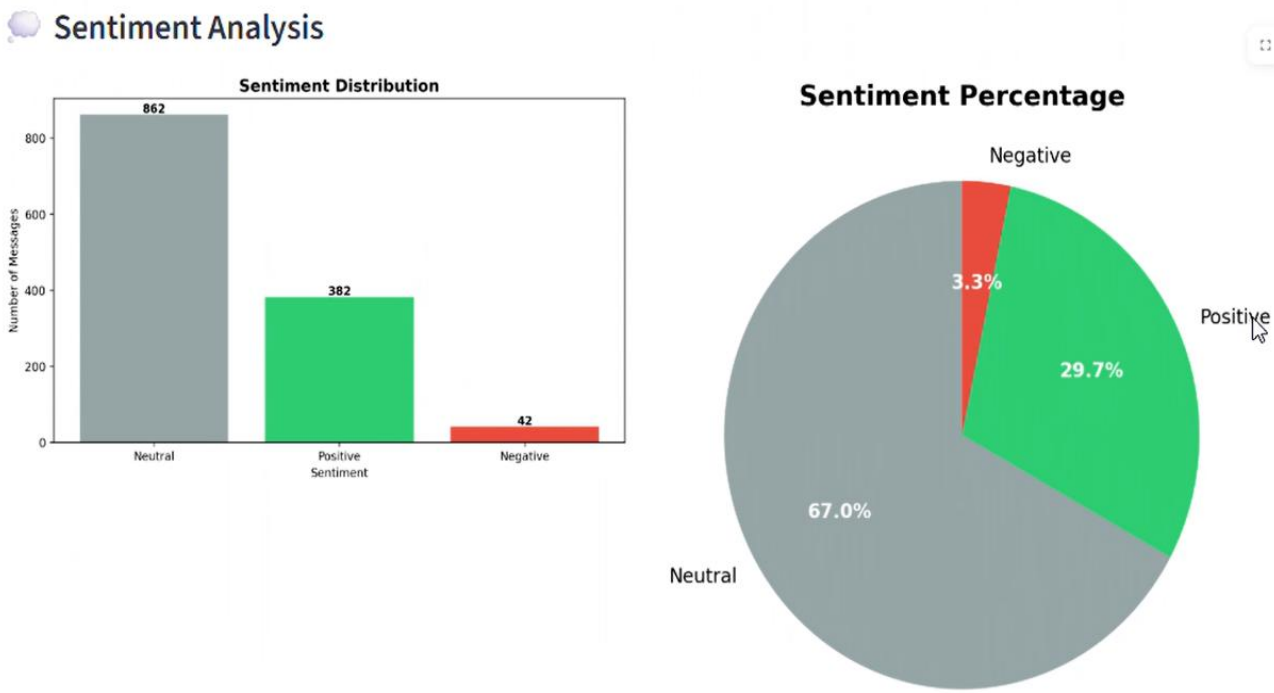


SAMPLE OUTPUT:

9.1 HOME SCREEN

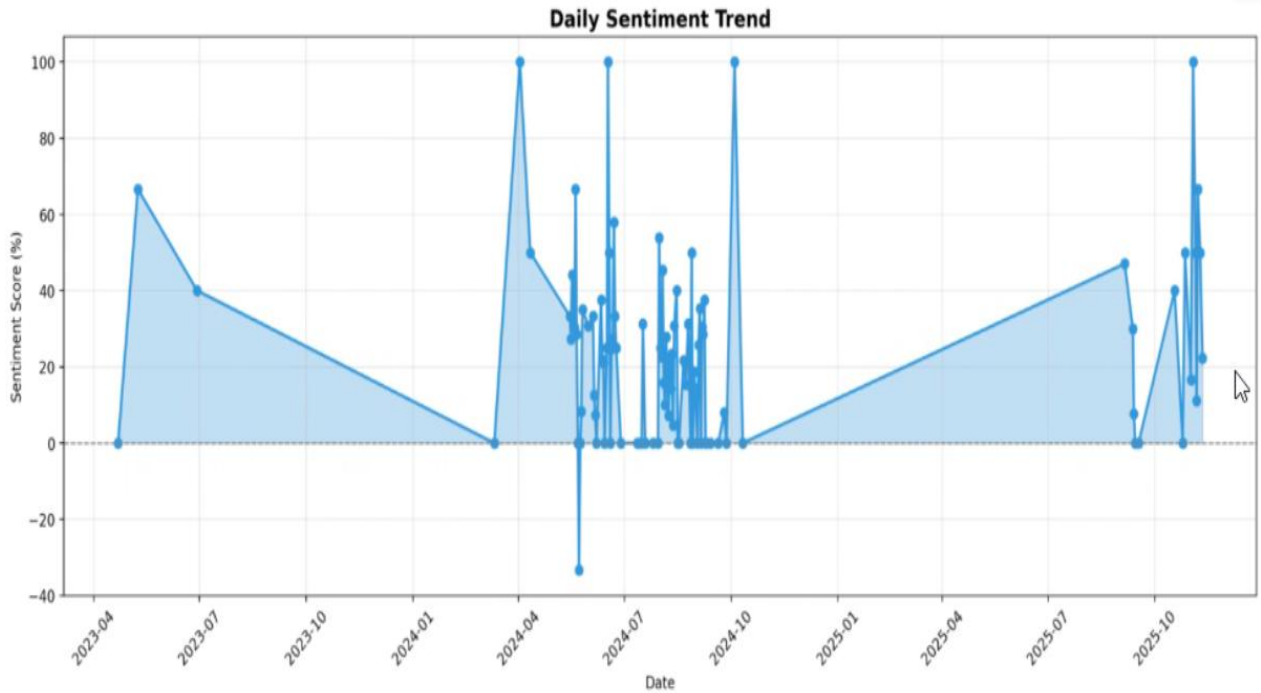


9.2 SENTIMENT DETECTED

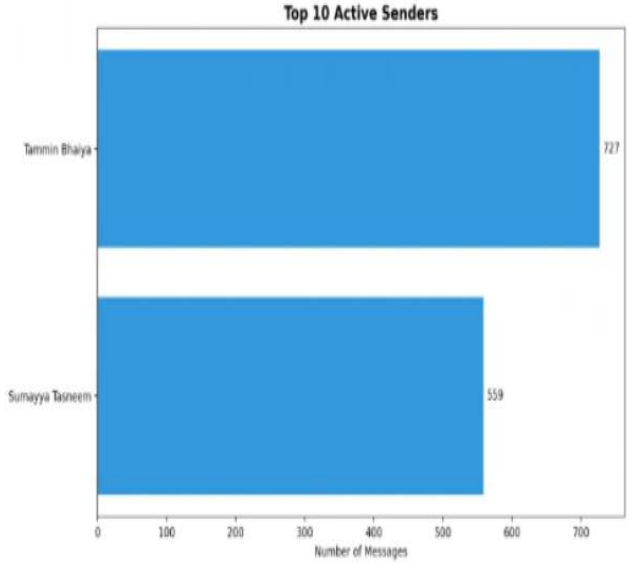


Deploy

📈 Sentiment Trend Over Time



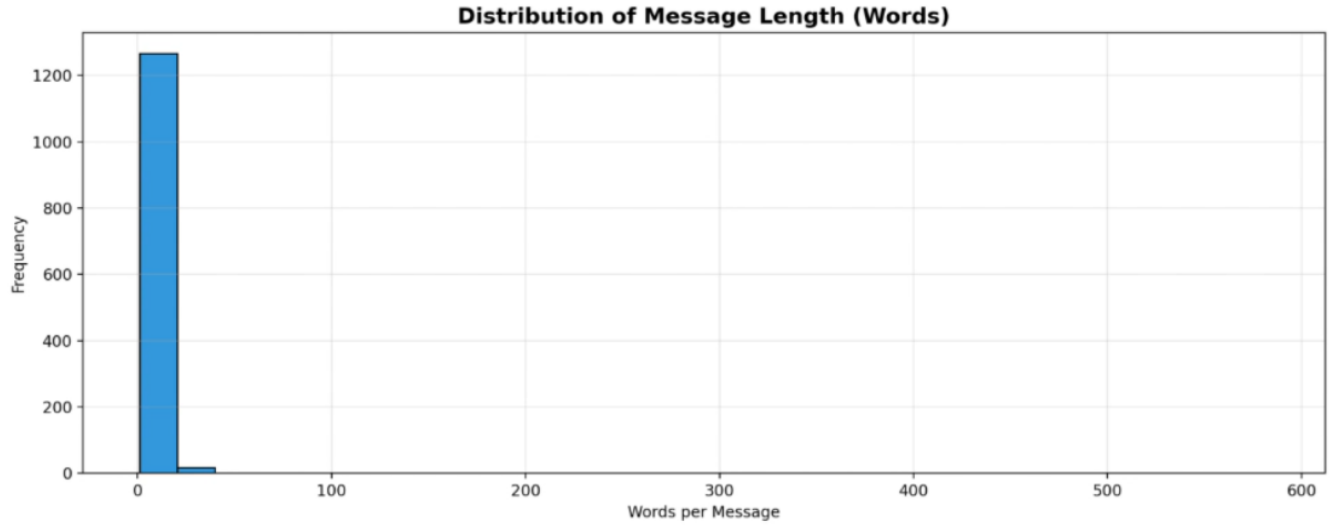
👥 Activity by Sender



sender	Messages	Avg Words	Positive Messages
Tammin Bhaiya	727	7.0605	234
Sumayya Tasneem	559	2.1574	148

Word Statistics

Total Words	Avg Words/Message	Max Words/Message
6339	4.9	583



Message Details

Filter by Sender

Tammin Bhaiya × Sumayya Tasneem ×

Filter by Sentiment

Positive × Negative × Neutral ×

	timestamp	sender	message	sentiment	word_count
1285	2025-11-11 19:56:00	Sumayya Tasneem	hehe	Neutral	1
1284	2025-11-11 19:26:00	Tammin Bhaiya	Sorry shines on u ✨	Negative	5
1283	2025-11-11 19:26:00	Tammin Bhaiya	Good… that smile looks nice on you take care	Positive	9
1282	2025-11-11 19:15:00	Sumayya Tasneem	😊	Neutral	1
1281	2025-11-11 19:12:00	Tammin Bhaiya	So stay happy… that's enough for me...	Positive	7
1280	2025-11-11 19:09:00	Tammin Bhaiya	Always anytime ur welcome Summu	Positive	5
1279	2025-11-11 19:05:00	Sumayya Tasneem	thanku bhaiya	Neutral	2
1278	2025-11-11 19:04:00	Sumayya Tasneem	haa	Neutral	1
1277	2025-11-11 18:59:00	Tammin Bhaiya	All the best dear ur always the best ✨	Positive	9
1276	2025-11-11 18:58:00	Tammin Bhaiya	Do it peacefully… Now summu project mode on	Positive	8

VERISPIRE